

SAMAY THAKUR

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EDUCATION

M.S. Ramaiah Institute of Technology	Oct 2023 – Present
Bachelor of Engineering Electronics and Instrumentation CGPA: 7.95	Bengaluru, India
Sunbeam School Varuna	2021 – 2022
Grade XII CBSE Percentage: 74.2%	Varanasi, India
Sunbeam School Varuna	2019 – 2020
Grade X CBSE Percentage: 89.6%	Varanasi, India

EXPERIENCE

Trainee	Aug 2024 – Aug 2024
CPS & IoT Intern (Intra Institutional Internship)	On-Site, India
• Developed a real-time water-level monitoring and automated pump-control system using Arduino Uno and ultrasonic sensors, implementing sensor data acquisition and threshold-based control logic in C/C++. • Tested and debugged the control system under varying water-level conditions to automate pump ON/OFF operation and prevent tank overflow and dry-run conditions.	

PROJECTS

AI-Based Bread Quality Indicator System GitHub
• Developed an end-to-end food-quality assessment solution combining MQ-135 gas sensors and ESP32-CAM image capture to enable real-time bread-quality evaluation. • Built a MobileNetV2-based image classification pipeline with 128×128 preprocessing, achieving 100% validation accuracy after 15 training epochs. • Designed a Flask-based monitoring interface integrating model predictions, Wi-Fi telemetry, and automated LED/buzzer alerts for real-time user feedback. • Technologies/Tools: Python, TensorFlow, MobileNetV2, TensorFlow Lite, Flask
Stock Market Price Prediction & Analysis GitHub
• Developed an LSTM-based forecasting solution to predict next-day NIFTY 50 price movements from historical data. • Engineered predictive features using RSI, MACD, Moving Averages, and Bollinger Bands to improve the model's ability to capture market trends. • Achieved 1.39% MAPE and 93%+ R^2 on next-day price prediction, evaluating performance against baseline forecasting models. • Technologies/Tools: Python, TensorFlow, LSTM, Pandas, NumPy, Scikit-learn

TECHNICAL SKILLS

Programming Languages: Python, C++
Concepts: Algorithms, Data Structures, Compiler Design, Computer Networks, Database Management Systems, Digital Logic Design, Operating System
Data & Analytics: Data Analysis, Data Visualization, Machine Learning
AI/ML: TensorFlow, TensorFlow Lite, Scikit-learn, Computer Vision, Deep Learning
Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, Flask
Soft Skills: Problem Solving, Communication, Teamwork, Leadership

ACHIEVEMENTS & LEADERSHIP

- **Student Member:** Instrument Society of India (ISOI), IISc Bangalore (Nov 2023 – Oct 2027)
- **Community Volunteer:** Organized and supported blood-donation drives through school and community initiatives, facilitating donor engagement and community welfare.